

BENNETT JACKSON

SOFTWARE ENGINEER

Tucson, AZ | (520) 869-9674 | BennettP.Jackson@gmail.com

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EDUCATION

University of Arizona | B.S. in Software Engineering (May 2025)

Aug 2021 - Present

GPA | 3.84

Awards | Dean's List with distinction (x3), Academic year distinction (x2), Engineering Honors Convocation Award (x2)

Extracurricular | Student advisory board for Software Engineering, Member of Software Engineering Wildcats Club

EXPERIENCE & PROJECTS

Software Engineer Intern | Garmin | *Chandler, AZ (On-site)*

May 2024 - Aug 2024

- Developed safety-critical embedded C software for several generations of Garmin Integrated Flight Decks, ensuring compliance with industry standards across 80+ aircraft models.
- Enhanced the testing process for 15 components by developing an automated regression testing suite in Python.
- Identified and resolved seven bugs found during testing through software and requirement changes.
- Implemented six customer-requested airframe features to enhance the functionality of Garmin Integrated Flight Decks.
- Leveraged software architecture to implement runtime-determined features, optimizing system flexibility.
- Utilized Git to manage and track changes, pushing 75+ commits to master during software development.

Student Software Developer | University of Arizona | *Tucson, AZ (On-site)*

Jan 2024 - May 2024

- Developed a full-stack web application for the Systems and Industrial Engineering Disruption Lab to convert and visualize Bayesian networks from OML, including a parser for efficient data translation, enhancing accessibility for researchers.
- Conducted thorough testing and debugging, achieving a 95% reduction in reported bugs during the testing phase.
- Built a Flask backend with six RESTful API endpoints and containerized the app using Docker for seamless deployment.
- Created a responsive frontend using HTML, CSS, and JavaScript, enhancing usability and satisfaction by 50%.

Software Engineer Intern | NASA Jet Propulsion Laboratory | *Pasadena, CA (Hybrid)*

Jan 2023 - Jan 2024

- Streamlined Python scripts with open-source libraries, Jama, and TestRail APIs, improving runtime by 32%.
- Led a team of three interns to develop an open-source software solution for an internal application migration.
- Implemented CI/CD pipelines with Jenkins for automated testing and deployment of two pip packages.
- Effectively managed tickets and metrics across multiple sprints using Jira, successfully completing 90+ tickets.
- Spearheaded internal customer-facing interactions to communicate project updates and address concerns.
- Utilized Git and GitHub for version control and collaboration, successfully merging 90+ pull requests to master.

Project | Weekly Playlist Manager

- Constructed in Python, this project leverages the Spotify API to automate playlist management, allowing users to retrieve and save songs seamlessly. Integrated CI/CD with GitHub Actions for regular updates and efficient deployment.

KEY SKILLS AND RELEVANT COURSEWORK

Coding Languages | C, C++, Python, Java, HTML, CSS, JavaScript, Verilog, Assembly, Bash, Swift

Technical Skills | Linux, Git and GitHub, CI/CD (Jenkins, GitHub Actions, Docker, Maven), Jira, SQL, Spring and Spring Boot, RESTful APIs, JUnit and Unit Testing, Software Security, Data Engineering and ETL, UML and OML, Flask, Pandas, RabbitMQ, Jama and TestRail, Scrum and Agile Methodologies, Object-Oriented Programming, Code Optimization, Command-Line Interface Open-source Practices, Anaconda

Additional Skills | Creative Solutions, Strategic Thinking, Teamwork, Communication, Software Development Lifecycle

Coursework | Software Architecture and Design, Software DevSecOps, Introduction to Cybersecurity Operations, Software Assurance and Security, Software Requirements Analysis, Software Project Management, Discrete Structures and Basic Algorithms, Computer Programming for Engineering Applications 1 and 2, Systems Cybersecurity Engineering, Fundamentals of Computer Organization, Discrete Math, Digital Logic, Interdisciplinary Capstone, Introduction to Web Design and Development, Linear Algebra, Engineering Ethics